
GazeBot

Head-Gaze and Sip-and-Puff Shared Autonomy System for Assistive Robot Manipulation

Using Hello Robot Stretch 3 and Meta Quest 3

Project Proposal

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Duration: 8 Weeks (10 hrs/week, 80 total hours)

Location: LIFE Home, 75 Bailey Drive, Champaign, IL

Executive Summary

This proposal outlines an 8-week project to develop **GazeBot**, a hands-free, voice-free teleoperation system that enables individuals with severe motor disabilities to control the Hello Robot Stretch 3 mobile manipulator using only head movements and a sip-and-puff switch via a Meta Quest 3 XR headset.

The target users—wheelchair-bound individuals with no hand function and no speech (e.g., advanced ALS, C1–C4 SCI, locked-in syndrome)—retain head and oral motor control, which GazeBot leverages as the sole inputs. A **shared autonomy paradigm** allows the user to provide high-level intent (selecting objects/destinations via head-gaze) while the robot autonomously handles navigation, grasp planning, and execution.

This directly aligns with the LIFE Home’s mission and builds on existing lab infrastructure (Stretch robot, Meta VR devices) and published work (Olatunji, Mahajan et al., 2024).

Literature Review

Assistive Robot Teleoperation

Olatunji et al. (2024) deployed a Hello Robot Stretch in the home of a quadriplegic older adult for 4 weeks through participatory design, achieving 68% performance increase and 72% satisfaction improvement. The UW CREATE lab demonstrated that customized interfaces significantly improve performance for users with disabilities. The ALS Association and University of Pittsburgh funded a \$400K project developing Stretch for people with ALS. Mahajan et al. (2025) further demonstrated robotic support for older adults with cognitive and mobility impairments using Stretch in the LIFE Home.

Gaze-Based Robot Control

MIHRaGe (2025) combines gaze tracking with mixed reality for hands-free object manipulation by mobility-impaired individuals. **RaycastGrasp (2025)** achieves >88% object detection confidence using stable gaze fixation. **Intent at a Glance (2026)** uses foundation models with gaze input for robotic intent inference. A scoping review (Frontiers, 2024) identified three approaches: telemanipulation with displays, directional gaze, and object-oriented gaze with 3D determination.

Shared Autonomy

Jeon et al. (RSS 2020) demonstrated learned latent actions enabling a 2-DoF joystick to control a 7-DoF robot arm—both faster and preferred by users. Admoni et al. (CMU, RSS 2015) blended brain-computer interface control with autonomous assistance, specifically exploring sip-and-puff for motor-impaired users. ASHA (2022) applied human-in-the-loop reinforcement learning for assistive teleoperation.

XR Robot Teleoperation

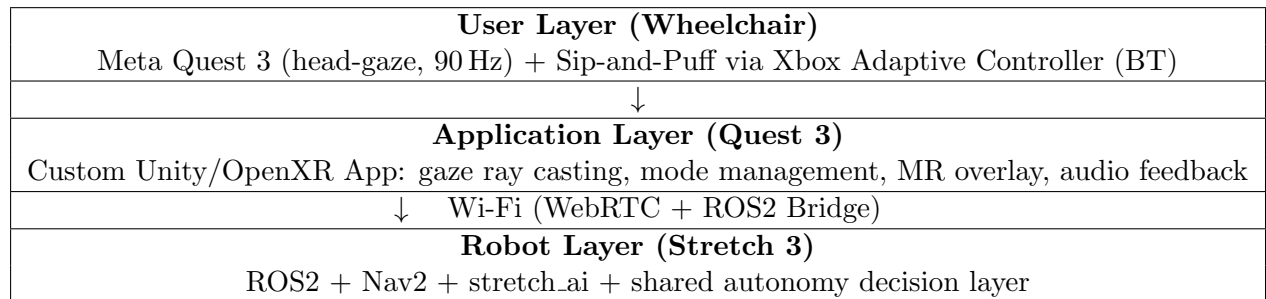
OPEN TEACH (CoRL 2024) provides open-source Meta Quest 3 teleoperation for Stretch at 90 Hz with 87% imitation learning success. **BEAVR (MIT, 2025)** achieves ≤ 35 ms latency VR teleoperation. **XRoboToolkit (2025)** offers a cross-platform OpenXR framework for robot control.

Gap in Literature

No existing system combines: (1) head-gaze control via XR headset, (2) sip-and-puff confirmation, (3) shared autonomy for grasp and navigation, and (4) the Stretch 3 manipulator—for users who cannot use hands or voice. **GazeBot fills this gap.**

System Architecture

Overview



Input Mapping

Input	Action
Head rotation (yaw/pitch)	Gaze ray direction—selects targets in robot camera view
Soft sip	Confirm / Select
Soft puff	Cancel / Back
Hard sip	Switch control mode
Hard puff	Emergency stop—immediately halts all robot motion

Control Modes

Mode	Head-Gaze Maps To	Sip Action
Look	Robot head pan/tilt (camera)	Set viewpoint
Navigate	Target destination in environment	Go to target (Nav2)
Manipulate	Object selection via gaze ray	Grasp/place (autonomous)
Menu	UI option navigation	Select preset task

Shared Autonomy Pipeline

1. **User looks at object** → head-gaze ray highlights detected object in MR overlay
2. **User sips to confirm** → `stretch_ai` identifies object via YOLO/SAM
3. **Robot plans grasp** → trajectory preview shown in MR view
4. **User sips to approve** → robot executes autonomous grasp
5. **User looks at destination** → head-gaze indicates placement location
6. **User sips to confirm** → robot navigates (Nav2) and places object

Technology Stack

Component	Technology
XR Application	Unity 2022 LTS + Meta XR SDK + OpenXR
Head Tracking	Meta Quest 3 inside-out tracking (6DoF, 90 Hz)
Sip-and-Puff	Generic switch + Xbox Adaptive Controller (Bluetooth)
Robot Control	ROS2 Humble + <code>stretch.body</code> Python SDK
Navigation	Nav2 (ROS2) with RPLIDAR A1
Object Detection	<code>stretch.ai</code> (YOLO + SAM)
Grasp Planning	<code>stretch.ai</code> / FUNMAP whole-body grasping
Video Streaming	WebRTC (via <code>stretch_web_teleop</code> infrastructure)
Audio Feedback	Bluetooth earbuds with confirmation tones

Methodology

The project follows an iterative, incremental development approach. Each week produces a functional system increment that can be demonstrated and tested in the LIFE Home environment.

Development workflow: Each component is built upon the previous week’s deliverable, starting with basic robot control and progressively adding head-gaze input, sip-and-puff integration, autonomous navigation, shared autonomy manipulation, and task presets.

Evaluation: Structured task trials conducted in the LIFE Home’s home simulation area with observation cameras recording for analysis. Five standard ADL tasks are evaluated across 3 trials each: (1) fetch a specified object, (2) navigate to a different room, (3) open a cabinet and retrieve an item, (4) pick up an object from the floor, (5) execute a preset “get water” sequence.

Instruments: NASA-TLX workload assessment and System Usability Scale (SUS), adapted for head-gaze input.

Timeline (8 Weeks)

Week	Phase	Key Tasks and Milestone
1	Setup & Baseline	Configure Stretch 3 with ROS2/ <code>stretch_ai</code> in LIFE Home. Set up Unity + Meta XR SDK. Get OPEN TEACH base-line running (Quest 3 → Stretch). <i>Milestone: Stretch controllable via Quest 3 hand tracking.</i>
2	Head-Gaze Control	Build Unity app reading head pose via OpenXR. Implement gaze ray casting. Stream Stretch camera into Quest 3 via WebRTC. Map head rotation to robot camera pan/tilt. <i>Milestone: User sees through robot's eyes, controls camera with head.</i>
3	SNP + Modes	Connect sip-and-puff to Xbox Adaptive Controller over Bluetooth. Map sip/puff signals to actions. Implement 4 control modes (Look, Navigate, Manipulate, Menu). Build MR HUD. <i>Milestone: Full hands-free input chain working across all modes.</i>
4	Navigation	Integrate Nav2—head-gaze selects target, sip confirms. Convert gaze ray + depth to 3D goal. Show path preview in MR. Test throughout LIFE Home. <i>Milestone: User looks at a point, sips, robot navigates there.</i>
5	Manipulation	Integrate <code>stretch_ai</code> object detection. Gaze ray highlights objects. Sip triggers autonomous grasp. Grasp preview in MR. “Bring to me” behavior. <i>Milestone: User looks at object, sips, robot picks up and brings it.</i>
6	Task Presets	Build preset tasks: “Fetch object”, “Open door”, “Clear path”, “Get water”. Head-gaze menu. Audio feedback via earbuds. Emergency stop refinement. <i>Milestone: Complete system with preset ADL tasks and safety stops.</i>
7	Evaluation	Structured evaluation: 5 tasks × 3 trials. Baseline comparison with <code>stretch_web_teleop</code> . Administer NASA-TLX and SUS. Data analysis. <i>Milestone: Quantitative evaluation data demonstrating viability.</i>
8	Dissemination	Record demo video (2–3 min) in LIFE Home. Write technical report. Clean codebase, push to GitHub. Prepare lab presentation. Draft conference paper outline. <i>Milestone: Demo video, report, GitHub repo, paper outline.</i>

Outcomes and Deliverables

1. **GazeBot Software System**—Open-source Unity application + ROS2 nodes enabling head-gaze + sip-and-puff control of Stretch 3 via Quest 3.
2. **Technical Report**—Detailed documentation of system architecture, implementation, and

evaluation results.

3. **Demo Video**—2–3 minute professional demonstration recorded in the LIFE Home showing the full workflow from a wheelchair user’s perspective.
4. **Evaluation Data**—Task performance data and subjective usability scores from structured trials in the LIFE Home.
5. **GitHub Repository**—Fully documented, reproducible codebase with setup instructions.
6. **Conference Paper Outline**—Draft targeting ACM ASSETS 2026, HRI 2027, or CHI 2027 Late-Breaking Work.

Required Resources

Available at LIFE Home

Resource	Status
Hello Robot Stretch (mobile manipulator)	Available (bookable)
Meta Quest / Oculus VR headset	Available in Extended Reality Lab
Home simulation area	Available for testing
Observation cameras (6× Axis M5525-E)	Available for evaluation recording
Wi-Fi infrastructure	Available

To Be Procured (Estimated: \$200–\$400)

Item	Est. Cost
Xbox Adaptive Controller	\$99
Sip-and-puff switch (3.5mm jack)	\$50–150
Gooseneck mount for SNP tube	\$20–40
Bluetooth earbuds (audio feedback)	\$30–50
Total	\$200–340

Software (All Free / Open-Source)

Unity 2022 LTS, Meta XR SDK, ROS2 Humble, `stretch_ai`, `stretch_body`, `stretch_ros2`, OPEN TEACH.

References

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